

Task 1: To Capture SIGTSTP, Suspend Processes, Update Job Table, Preserve State, Resume Later, Test Suspension Workflow

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

pid_t child_pid = -1;

void sigtstp_handler(int sig) {
    printf("\nSIGTSTP received.\n");

    if (child_pid > 0) {
        printf("Suspending foreground job: %d\n", child_pid);
        kill(child_pid, SIGSTOP);
    }
}

int main() {
    struct sigaction sa;
    int status;

    sa.sa_handler = sigtstp_handler;
    sigemptyset(&sa.sa_mask);
    sa.sa_flags = 0;

    sigaction(SIGTSTP, &sa, NULL);
```

```
        sleep(1);
    }

    printf("\nJob can now be resumed using SIGCONT.\n");
    sleep(3);
    printf("Sending SIGCONT...\n");
    kill(child_pid, SIGCONT);
    sleep(5);
    printf("Terminating resumed job...\n");
    kill(child_pid, SIGTERM);
    waitpid(child_pid, &status, 0);
    printf("Job completed.\n");
    return 0;
}
```

```
shreyas@Grigori:~$ ps -o pid,ppid,pgid,state,cmd
  PID    PPID    PGID S  CMD
  1279    1273    1279 S  -bash
  1295    1279    1295 R  ps -o pid,ppid,pgid,state,cmd
shreyas@Grigori:~$ pgrep task1
shreyas@Grigori:~$ pgrep task1
shreyas@Grigori:~$ |
```

```
cd ~/sigtstp_tasks
shreyas@Grigori:~/sigtstp_tasks$ nano task1.c
shreyas@Grigori:~/sigtstp_tasks$ mkdir -p ~/sigtstp_tasks
cd ~/sigtstp_tasks
shreyas@Grigori:~/sigtstp_tasks$ gcc task1.c -o task1
shreyas@Grigori:~/sigtstp_tasks$ ./task1
Foreground job PID = 1271
Press Ctrl+Z to suspend the job.
Child process started. PID = 1271
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
Job running...
^Z
SIGTSTP received.
Suspending foreground job: 1271

Job suspended.
Job state preserved.

Job can now be resumed using SIGCONT.
Sending SIGCONT...
Job running...
Job running...
Job running...
Terminating resumed job...
```

Task 2: To Create Process Groups, Assign Group IDs, Transfer Terminal Control, Restore Ownership, Coordinate Signals, Validate Behavior

```
shreyas@Grigori:~$ nano task2.c
shreyas@Grigori:~$ gcc task2.c -o task2
shreyas@Grigori:~$ ./task2
Shell PID = 1344
Shell PGID = 1344
Child 1 PID = 1345
Child 1 PGID = 1345
Child 1 running...

Process group created.
Group ID = 1345

Transferring terminal control to process group...
Child 2 PID = 1346
Child 2 PGID = 1345
Child 2 running...
Child 1 running...
Child 2 running...
Child 1 running...
Child 2 running...

Sending SIGINT to entire process group...
Restoring terminal ownership to shell...

[1]+  Stopped                  ./task2
shreyas@Grigori:~$ ps -o pid,ppid,pgid,state,cmd
  PID   PPID   PGID S  CMD
  1318   1314   1318 S  -bash
  1344   1318   1344 T  ./task2
  1345   1344   1345 Z  [task2] <defunct>
```

```
  1344   1318   1344 T  ./task2
  1345   1344   1345 Z  [task2] <defunct>
  1346   1344   1345 Z  [task2] <defunct>
  1372   1318   1372 R  ps -o pid,ppid,pgid,state,cmd
shreyas@Grigori:~$ ps -o pid,ppid,pgid,state,cmd
```

```
shreyas@Grigori:~$ ps -o pid,ppid,pgid,sid,tpgid,state,cmd
  PID   PPID   PGID   SID   TPGID S  CMD
  1353   1348   1353   1353   1369 S  -bash
  1369   1353   1369   1353   1369 R  ps -o pid,ppid,pgid,sid,tpgid,state,cmd
shreyas@Grigori:~$ pgrep task2
1344
1345
1346
shreyas@Grigori:~$ ps -o pid,pgid,tpgid,cmd
  PID   PGID   TPGID CMD
  1353   1353   1371 -bash
  1371   1371   1371 ps -o pid,pgid,tpgid,cmd
shreyas@Grigori:~$ kill -PGID
-bash: kill: PGID: invalid signal specification
shreyas@Grigori:~$ kill 1353
shreyas@Grigori:~$ kill 1371
-bash: kill: (1371) - No such process
shreyas@Grigori:~$ kill(-child1, SIGINT);
-bash: syntax error near unexpected token `~child1,'
shreyas@Grigori:~$ |
```